0.1 Sylow's Theorem

> **Theorem 0.1.1.** *Sylow's Theorem*
> Let G be a group of order $p^{\alpha}m$, where p is a prime such that $p \nmid m$.
> A group of order $p^r,\ (r \geq 1)$ is called a p-group, Subgroups of G which are p-groups are called p-subgroup.
> In particular, subgroups of order p^{α} is called *Sylow p-subgroup* of G. And, define a collection
>
> $$\mathtt{Syl}_p(G) \stackrel{\text{def}}{=} \{P \leq G \mid |P| = p^{\alpha}\},\ n_p(G) \stackrel{\text{def}}{=} \mathtt{Card}(\mathtt{Syl}_p(G))$$
>
> *The First Sylow Theorem*
> There exists a Sylow p-subgroup of G. i.e., $\mathtt{Syl}_p(G) \neq \emptyset$.
>
> *The Second Sylow Theorem*
> If $P \in \mathtt{Syl}_p(G)$ and $Q \leq G$ be a p-subgroup. Then, there exists $g \in G$ such that $Q \leq gPg^{-1}$.
>
> *The Third Sylow Theorem*
> $n_p \equiv 1 (\mathrm{mod} p)$, $n_p = |G : N_G(P)|$ for any $P \in \mathtt{Syl}_p(G)$, and $n_p \mid m$.

Before prove above statments, we show that:

Lemma 0.1.2. Let $P \in \mathtt{Syl}_p(G)$. If Q is p-subgroup of G, then $Q \cap N_G(P) = Q \cap P$.

Proof. Put $H = Q \cap N_G(P)$. Since $P \leq G$, for any $p \in P$, $pPp^{-1} = P$, thus $p \in N_G(P)$. i.e., $P \leq N_G(P)$.
Thus, Enough to Show that $H \leq Q \cap P$. Since $H \leq N_G(P)$,

$$PH = \bigcup_{h \in H} Ph = \bigcup_{h \in H} hP = HP$$

Thus, $PH \leq G$. And,

$$|PH| = \frac{|P||H|}{|P \cap H|}$$

By Lagrange's Theorem, $H \leq P$ and $P \cap H \leq P$ must have order of powers of p, so PH be a p-group. Clearly, $P \leq PH$ and P is the largest p-group of G, thus, $PH = P$. This means, $H \leq P$. □

Proof. The First Theorem: The existence of Sylow p-subgroup. Proof by Induction:
If $|G| = 1$, there is nothing to prove.
Assume inductively the existence of Sylow p-subgroups for all groups of order less than $|G|$.
In case of $p||Z(G)|$, then by Cauchy's Theorem, $Z(G)$ has a subgroup N which has order of p.
Clearly N is Normal, and $G/N = |G|/|N| = p^{a-1}m$. By assumption, G/N has a subgroup P' of order $p^{\alpha-1}$.
By The Forth Isomorphism Theorem, Let $P \leq G$ be a subgroup such that $P/N = P'$.
Then, $|P| = |P/N| \cdot |N| = p^{\alpha}$, Thus P be a Sylow p-subgroup of G.
In case of $p \nmid |Z(G)|$.
Let $g_1, \ldots, g_r$ be represectatives of the distinct conjugacy classes of G, not contained in $Z(G)$. Then, The Class Equation gives

$$|G| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|$$

Since p divides $|G|$, if for all $i = 1, 2, \ldots, r$, $p \mid |G : C_G(g_i)|$ then $p \mid |Z(G)|$, this is contradiction.
Thus, for some j, $p \nmid |G : C_G(g_j)|$. Put $H = C_G(g_j) < G$. Then, $|H|$ has a factor of p^{α}, by $p \nmid |G : C_G(g_j)|$. Now,

$$|H| = p^{\alpha}m' \quad (m' < m)$$

By assumption, H has a Sylow p-group, order of p^{α}.
Consequently, the existence of Sylow p-subgroup was shown.
The Second Theorem: Relation of p-subgroups.
The First Theorem gives existence of Sylow p-subgroups. Let $P \in \mathtt{Syl}_p(G)$. Denote that:

$$S \stackrel{\text{def}}{=} \{gPg^{-1} \mid g \in G\} = \{P_1, \ldots, P_r\}$$

Let $Q \leq G$ be an any p-subgroup of G. And, Q acts by conjucation on S. i.e.,

$$\alpha : Q \times S \to S : (q, P_i) \mapsto qP_iq^{-1}$$

Write S as a disjoint union of orbits under this action by Q:

$$S = \mathcal{O}_1 \cup \mathcal{O}_2 \cup \cdots \cup \mathcal{O}_s$$

where $r = |\mathcal{O}_1| + \cdots + |\mathcal{O}_s|$. Rearrange a set S as: $P_i \in \mathcal{O}_i,\ 1 \leq i \leq s$. Now, using Definition, Lemma, and above Theorem,

$$|\mathcal{O}_i| \overset{\text{Thm}}{=} |Q : N_Q(P_i)| \overset{\text{def}}{=} |Q : N_G(P_i) \cap Q| \overset{\text{lemma}}{=} |Q : P_i \cap Q|$$

for each $1 \leq i \leq s$. Since Q was arbitrary, Let $Q = P_1$, so that $|\mathcal{O}_1| = |P_1 : P_1 \cap P_1| = 1$. And, for each $i \geq 2$, $P_i \cap P_1 < P_1$,

$$|\mathcal{O}_i| = |P_1 : P_i \cap P_1| > 1$$

Since $P_1 \in \mathrm{Syl}_p(G)$, that is $|P_1| = p^\alpha$, $|P_1 : P_i \cap P_1| = |P_1|/|P_i \cap P_1| = p^k$ where $1 \leq k < \alpha$.
This means for each $2 \leq i \leq s$, p divides $|\mathcal{O}_i|$. Thus,

$$r = |\mathcal{O}_1| + (|\mathcal{O}_2| + \cdots + |\mathcal{O}_s|) \equiv 1 (\mathrm{mod} p)$$

Now, Proof by Contradiction: Let $Q \leq G$ be a p-subgroup. Suppose that for any $1 \leq i \leq r$, $Q \nleq P_i$.
Then, $P_i \cap Q < Q$ for all i, this means

$$|\mathcal{O}_i| = |Q : P_i \cap Q| > 1$$

Thus for any i, p divides $|\mathcal{O}_i|$, this is Contradiction. This proved Relation of p-subgroups.
Finally, The Third Theorem:
Since Second Theorem, this gives that $S = \mathrm{Syl}_p(G)$, thus $n_p(G) = r$. That is, $n_p \equiv 1 (\mathrm{mod} p)$.
Since all Sylow p-subgroups are Conjugate, for any $P \in \mathrm{Syl}_p(G)$,

$$n_p = r = |\mathcal{O}_1| = |G : N_G(P)|$$

Consequently, Completing the Sylow Theorem. □

Corollary 0.1.3. Let P be a Sylow p-subgroup of G. Then, TFAE:

1. P is the unique Sylow p-subgroup of G. That is, $n_p = 1$.
2. P is Normal in G.
3. P is characteristic in G.
4. If $X \subset G$ satisfies for all $x \in X$, order of x is power of p, then $\langle X \rangle$ is a p-group.

0.2 Examples

0.2.1 Order of pq, p and q are primes with $p < q$

0.2.2 Order of pqr, p, q and r are primes with $p < q < r$

0.2.3 Order of p^2q, p and q are distinct primes